Selection, Maintenance and Gravitational Response: Modular Constraints and a Physical Return

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Abstract

Can the physical selection of observable algebras account for gravitational dynamics, rather than merely change what an observer can reconstruct? We formulate this question as an adiabatic evolution of selected localization structures and a common-action consistency requirement. Modular stability alone does not select the algebra. A supplied half-sided modular family and holographic dictionary give a conditional linearized gravity route. In the same-spacetime conformal scalar model of Consciousness II, the improved stress and its Wald surface coefficient connect accessible preparations to an all-ball test of the Einstein response. The test still requires modular consistency; changing a stress-containing algebra at fixed matter history does not change its first-order sourced Einstein tensor.

The source side is constructive. Calibrated correction, selected projections and learned protection change stress through actual interactions. In a specified Jackiw–Teitelboim theory, an acquired quantum record drives a material force and radiation, whose response changes later operations. The finite mean-action return retains old references, clock energy, support work and recoil. Separate spatial and full-mode bulk constructions extend the source account without identifying their distinct models.

Maintenance supplies a possible selection pressure. For complete entropy removal at rate $\dot{s}$ through a thermal chiral exhaust of fixed capacity c_{ex}, the heat rate obeys $\dot{Q} \geq k_B T \dot{s} + 3\hbar \dot{s}^2/(\pi c_{\mathrm{ex}})$. Pointer alignment lowers the complete renewal debit, and matched horizon transport reads its price. Sustained renewal and joint sharp exhaust with persistent causal return remain open. The proposed gravitational mechanism must connect formed access, a conserved total source and geometric consistency in one dynamics; neither a visibility filter nor a finite response in an assigned gravity theory supplies that connection.

Technical appendices and their original numbering are retained in HAFF-D technical supplement (`HAFF_D_supplement.pdf`). The certificate guide at the end of this paper identifies their claims.

1 The problem: what makes selection gravitational?

An apparatus changes which distinctions it can preserve and which operations it can execute. This can reorganize a reconstructed geometry. It can also change matter through

the selecting interaction. A theory of gravity based on observable-algebra selection must connect these two effects: the localization structure selected by physical histories must constrain a geometric law, and the same histories must supply the source and respond to it.

The central proposal is that gravitational dynamics describes a slow evolution of the selected geometric algebra, not just motion of matter within a fixed localization structure. This is a mechanism conjecture, developed in Section 4, rather than an identification of every algebra change with curvature. Its strongest obstruction is already visible in a conformal model: two selected algebras that retain the same stress readings can give different relative entropies while fixing the same matter-sourced Einstein tensor. A physical change of localization must therefore be distinguished from a change of description.

HAFF-A, B and E relate channels, stable content and actual records [19, 20]. Foundation I constructs a stable hull relative to a readable seed [54]. These results make parts of algebra selection precise; they do not determine a spacetime metric or a universal gravitational dynamics. Here we ask what further relation would make that selection gravitational.

Sections 2–5 state the algebraic question, its central conjecture and the modular constraints. The holographic route needs an explicit region family and dictionary. Section 6 gives a different conditional bridge in the same spacetime: one conformal scalar source, its surface charge, and its all-ball geometric test. That bridge identifies which law is compatible with the supplied consistency condition, while leaving the physical origin of that condition to the common-action conjecture.

Sections 8 and 9 then test the source and return edges through actual interactions in a specified two-dimensional JT theory. The decisive finite mechanism is an acquired record that changes a material source, not a command label attached to a separate emitter. Section 10 prices the full maintained state. Section 11.2 asks whether the source, renewal and informative return can persist together. Long controller, clock, support and source proofs are in the appendices.

Table 1 separates these constructions. Its rows are not pieces of one already realized apparatus. In particular, a supplied JT response cannot establish the higher-dimensional consistency law, and a modular constraint cannot manufacture a controller.

2 The Factorization Problem

2.1 No canonical tensor factorization

A Hilbert space of dimension $d = d_1 d_2$ admits a tensor factorization, but its Hilbert structure alone does not select one. Commuting observable algebras can induce the subsystem structure [17, 18]. Which of those algebras an apparatus can use is a physical question about its couplings and resources.

2.2 Coarse-graining and effective descriptions

Paper A [19] gives the following conditional separation.

Proposition 2.1 (Conditional coarse-graining separation)**.** *Let $\Phi_1 \neq \Phi_2$ be finite-dimensional coarse-graining channels with a common output space. Their outputs differ for generic pure inputs. Entropy profiles differ only under the additional nondegeneracy*

Object and supplied inputs	Result	Remaining interface
Modular family, dictionary and ball laws	Conditional translations and linearized geometric consistency (Sections 2–5)	Formation of the candidate family and its geometric dictionary.
Conformal scalar, source reference, collar prescription and MC	One prepared source, matched Wald coefficient and all-ball Einstein test (Section 6)	Physical origin of MC; apparatus source and finite-resource readings.
Calibrated field operations and a JT action	Derived outgoing source, retarded response and finite echo/contact readout (Section 8; Appendix G)	Total apparatus stress and physical references; the action is an input.
Quantum core with classical mechanics and mean-action source	Finite retained-history return and strong-domain perturbation bounds (Section 9; Foundation III)	Fixed-window mean-action result; protected content, old references and initial work remain charged. No quantum stress tensor for the scalar fields or support is supplied.
Spatial quadratic fields and real-quartet bulk	Work-corrected full-mode stress matching, a finite spatial return, and a separately constrained quartet interior	Joining their receivers, moving endpoints, preparation and thermal ports in one material model.
Complete retained entropy and paid finite carrier	Long-window heat floor and neutral-sector sorting of a resettable source (Section 10; Proposition 11.2)	Actual adaptive source concentration, protected-content compatibility and local delayed readout.
One reused body over growing runs	Conjecture C.1 jointly requires complete-output sharpness, normalized window contrast and causal transition gain	Sustained inlet, support, sensing and resource account in that same dynamics; none is supplied by sorting alone.

Table 1: Objects, results and interfaces. Finite constructions retain their stated inputs; the joint growing-window target is a conjecture.

hypothesis stated in Paper A, and metric inequivalence follows only when a supplied reconstruction dictionary separates those profiles modulo diffeomorphisms.

Channel inequivalence therefore establishes a difference of effective descriptions, not by itself a difference of spacetime metrics. Paper A also decides, for qubit reductions, exactly when two channels give the same output entropy on every pure input, and it constructs one global state, one family of local noise channels and one fixed reconstruction rule that yield non-isometric finite metrics. Metric inequivalence of spacetime geometries remains a conditional model claim throughout this paper.

2.3 Geometry from entanglement

In a holographic model the Ryu–Takayanagi formula [27] relates an extremal area to boundary entropy,

$$S_A = \frac{\mathrm{Area}(\gamma_A)}{4G_N\hbar}, \tag{1}$$

and motivates an entanglement account of connectivity [28]. Such a reconstruction requires its dictionary. Outside a specified model, the accessible data determine operator relations rather than a spacetime metric.

3 A Structural Distinction: Gauge Forces and Gravity

Gauge interactions evolve matter and connection fields within a supplied localization structure. The algebraic proposal asks whether gravity instead describes how physical correlations, accessible operations and their stability select that structure. With a geometric dictionary, changing those relations can change effective distances and causal structure. The source constructions below test a necessary interface: changed access reaches geometry through a specified matter process.

If the proposal holds, quantum gravity must describe selection of localization as well as fields within it. This is not an explanation of why quantization fails. Background independence [29, 30] asks for covariance under changes of selected geometric structure, not equality of the metrics assigned to operationally different algebras. Selection is constrained by interactions and stability, rather than subjective choice [20]; changed knowledge affects stress only through an operation.

A common reconstruction dictionary gives every matter representation in one selected geometric sector the same metric. This conditional universality is not a derivation of free fall or equality of inertial and gravitational mass. Degrees outside the sector cannot be declared geometrically irrelevant while changing its metric: a dark-sector application would need an enlarged source algebra and explicit backreaction map. No dark-matter or dark-energy conclusion follows here.

4 Algebra selection as a gravitational mechanism

4.1 Setup

We separate the operator-algebraic and finite-dimensional regimes result by result. Modular statements use a Type III_1 ambient factor with a faithful normal state and explicit cyclic and separating hypotheses. Density matrices, traces, the quantum Fisher information metric and the qubit chain are finite-dimensional statements. The Rindler example is an algebraic quantum field theory check. Theorem 5.6 uses the holographic regime; Section 6 uses a distinct same-spacetime conformal model. The JT model of Section 8 is a supplied two-dimensional gravity theory.

Definition 4.1 (Selected geometric algebra)**.** *A* ***selected geometric algebra*** $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ *is a von Neumann subalgebra chosen as input to a specified algebra-to-geometry reconstruction model. It is not Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set. A coarse-graining model must declare the channel or inclusion relating any of those objects to* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$*.*

No canonical tensor factorization is assumed. State restrictions and relative-entropy data on $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ can be associated, as in holography and tensor networks, with an effective distance structure. That structure depends on the choice of $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$, on the stability of correlations under coarse-grained dynamics and on the redundancy of the encoding; it is an effective description valid in the context the algebra defines.

4.2 Central conjecture

Conjecture 4.2 (Gravity as adiabatic algebra evolution)**.** *Gravitational dynamics and the* ***adiabatic flow*** *of selected geometric algebras* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t)$ *arise from one physical selection process:*

1. *the global state evolves unitarily,* $|\Psi_U(t)\rangle = U(t)|\Psi_U(0)\rangle$*;*
2. *a model-dependent rule selects a geometric candidate* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \subset \mathcal{M}_U$ *at each time;*
3. *as the state evolves, the selected candidate may shift,* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \to \mathcal{A}_{\mathrm{geo},\mathbf{c}}(t+dt)$*;*
4. *the interaction that changes the selected algebra also determines the complete physical source and the compatible localization data. A change of algebra alone, with the matter history and source readings held fixed, supplies no additional first-order sourced Einstein tensor in the common-source regime below. A changed sourced response must be carried by the changed total source under that regime's field-law conditions.*

The proposed distinction from ordinary matter feedback is that the same dynamics selects the localization structure in which the source and response are represented, rather than merely evolving matter on a permanently supplied stage. Evolution within a fixed selected localization can describe gauge dynamics, but neither unitarity nor a moving algebra by itself identifies an interaction as gauge or gravitational. The selection claim must be established through the common-action conditions below; finite sourced return alone does not establish it.

The proposed flow needs a physical selector. A readable history changes which controls and correlations remain stable; those controls determine a selected hull and the matter interactions that sustain it. The changing hull must then admit compatible geometric region data, rather than merely receive an arbitrary distance assignment. In an adiabatic sector, the relaxation time of the selected structure must be short compared with its change time. This separation is a condition to establish in a model, not a consequence of writing $\mathcal{A}(t)$.

Two explanations compete here. In the reconstruction explanation, the ambient geometry and complete source stay fixed, while smaller algebras resolve less of them. In the dynamical explanation, history changes an actual coupling or preparation, changes the complete source, and changes later physical readings. The latter has a finite realization below; the stronger claim that the localization law itself is selected requires Section 7. Algebra variation alone distinguishes neither explanation.

A useful test compares histories under one action and one source calibration. Erasing command access before execution should remove the record-mediated part of the source; erasing a log after execution should not. If two selected structures contain the same total stress and surface data, a first-order difference of their sourced Einstein tensors would contradict the conformal bridge in Section 6. A difference only in relative entropy supports a reconstruction claim unless a corresponding physical intervention is demonstrated. Conversely, a history-dependent source that fails the same all-ball response test does not realize the proposed gravitational flow, even if it produces a detectable JT signal. Rapid algebra transitions lie outside the adiabatic claim; whether they admit a different geometric regime is open.

4.3 Comparison with quantum information geometry

For comparison consider the quantum Fisher information metric (QFIM) on a smooth finite-dimensional family of faithful effective states $\rho(\lambda)$ [36, 37],

$$g_{\mu\nu}(\lambda) = \frac{1}{2}\mathrm{Tr}\left(\rho(\lambda)\{L_\mu, L_\nu\}\right), \qquad \partial_\mu\rho = \frac{1}{2}(\rho L_\mu + L_\mu\rho). \tag{2}$$

It is coordinate independent, reduces to the classical Fisher metric in commuting limits, and measures how hard nearby states are to distinguish. It is also positive semidefinite, whereas a Lorentzian spacetime metric is indefinite and carries causal structure. Nothing in Eq. (2) supplies the negative direction, so the QFIM is a metric on model space and an analogy, not the spacetime metric. A change $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \to \mathcal{A}_{\mathrm{geo},\mathbf{c}} + \delta\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ may change this state-space metric; calling the change a spacetime perturbation requires an additional Lorentzian reconstruction map. In AdS/CFT, relative entropy and the entanglement first law constrain bulk perturbations [38, 39]; those results use the holographic dictionary and do not make an arbitrary boundary QFIM the bulk metric.

5 Algebraic Foundation: Modular-Stable Candidates

This section places one candidate class in Tomita–Takesaki modular theory. It does not translate measurement accessibility or environmental redundancy into modular conditions, and it does not select a unique algebra.

Standard modular definitions and their two checks are in Appendix E.

5.1 Modular definition of candidate algebras

Let $\mathcal{M}_U$ be a Type III_1 factor in standard form on $\mathcal{H}_U$, let $|\Psi_U\rangle$ be cyclic and separating, and write $\omega(\cdot) = \langle\Psi_U| \cdot |\Psi_U\rangle$. Then ω is faithful and normal and its ambient modular group $\sigma_t^{\omega,U}$ exists. Let $\{\mathcal{E}_t\}_{t\geq 0}$ be a normal unital completely positive (UCP) semigroup acting in the Heisenberg picture on $\mathcal{M}_U$.

Definition 5.1 (Modular-stable geometric candidate)**.** *A von Neumann subalgebra $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \subset \mathcal{M}_U$ is a* ***modular-stable geometric candidate*** *if:*

(U1) ***Faithful restriction.*** *$\omega|_{\mathcal{A}_{\mathrm{geo},\mathbf{c}}}$ is faithful and normal. In the faithful ambient setting this holds for every unital subalgebra, so it is a regularity condition, not a selector.*

(U2) ***Ambient modular stability.*** *$\sigma_t^{\omega,U}(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) = \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all real t.*

(U3) ***Semigroup stability.*** *$\mathcal{E}_t(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) \subseteq \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all $t \geq 0$.*

Remark 5.2 (Typed relation to the measurement objects)**.** *Conditions (U1)–(U3) do not derive Paper E's implementable effects* Eff*, their operator system* $\mathrm{Eff}_{\mathrm{span}}$ *or the pointer algebra $\mathcal{P}_{\mathrm{env}}$, and setwise semigroup stability is not the exact fixed set* $\mathrm{Fix}(\mathcal{E})$*. A downstream model may specify a channel or inclusion from those objects into $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$; without it the relation is motivational. Environmental redundancy does not imply faithfulness of a state restriction. A causal restriction map and a Lindblad evolution are different types, and each model must supply its own $\mathcal{E}_t$ and verify invariance.*

5.2 Selection is not settled by these conditions

Remark 5.3 (No uniqueness from (U1)–(U3))**.** *Conditions (U1)–(U3), even with only scalars fixed by both dynamics, do not determine a unique candidate. There are four independent obstructions.*

1. *Uniqueness of the modular flow for a fixed pair $(\mathcal{A}, \omega)$ does not select the algebra $\mathcal{A}$.*

2. *A normal completely positive map may preserve two subalgebras separately without preserving their join, because it is not multiplicative. Join preservation holds for modular automorphisms but not for a general $\mathcal{E}_t$; it does hold for subalgebras of the fixed algebra of a state-preserving dynamics, on which $\mathcal{E}_t$ acts as the identity [54].*

3. *Pointwise ergodicity constrains fixed elements, while (U2)–(U3) are setwise conditions; no implication to a unique maximal element of the lattice was proved.*

4. *With a faithful ambient state, $\mathcal{M}_U$ itself satisfies (U1)–(U3), so a bare maximality rule selects the ambient algebra unless an independent seed, properness, read-out or minimal-hull condition is supplied.*

Replacing $\mathcal{B}(\mathcal{H}_U)$ by a faithful Type III factor repairs the existence of the ambient modular flow but none of these selection failures. The class of candidates may contain several members that are not unitarily equivalent, and classifying it requires additional structure. A nontrivial selection principle could start from a foreground seed and construct a minimal invariant hull, or impose an explicit read-out constraint. Foundation I carries out the first

route for a supplied seed: under state-preserving dynamics the hull of a seed in the fixed algebra lies between that seed and the fixed algebra. For a minimal compact symmetry, it equals the seed-stabilizer fixed algebra when that algebra is preserved by the hull generators and the hull contains the group-fixed algebra [54]. These hypotheses establish uniqueness relative to the seed; the seed's physical origin is a further question.

5.3 Connection to linearized gravity

Theorem 5.4 (Entanglement first law in a density-matrix setting)**.** *Let ρ_0 be a faithful finite-dimensional density matrix and $\rho_\varepsilon = \rho_0 + \varepsilon\,\delta\rho + O(\varepsilon^2)$ a normalized perturbation, with $K_0 = -\log\rho_0$. Then $\delta S = \delta\langle K_0\rangle$. An infinite-dimensional extension requires entropy differentiability and control of the modular-energy variation; faithfulness alone is insufficient. For a Type III local algebra the statement must instead use Araki relative entropy.*

Proof sketch. $S(\rho_\varepsilon\|\rho_0) = -S(\rho_\varepsilon) + \langle K_0\rangle_{\rho_\varepsilon}$ plus a constant is nonnegative and stationary at $\varepsilon = 0$. □

Conjecture 4.2 posits a family $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(\lambda)$. Modular theory treats two kinds of perturbation differently. For a state perturbation on a fixed algebra the Connes cocycle $u_t = (D\omega' : D\omega)_t$ relates the automorphism groups by $\sigma_t^{\omega'}(A) = u_t\sigma_t^{\omega}(A)u_t^*$ [50, 40]. This is not a left-multiplication formula for the two full modular operators. If the algebra family is generated by the positive translation of an HSMI, $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(a) = e^{iaP}\mathcal{A}_{\mathrm{geo},\mathbf{c}}(0)e^{-iaP}$, and the vacuum is fixed by this translation, its full modular generator obeys $K(a) = e^{iaP}K(0)e^{-iaP}$. On a common smooth domain,

$$\frac{dK(a)}{da}\Big|_{a=0} = i[P, K(0)]. \tag{3}$$

Equation (3) concerns an HSMI-generated family and a derivative of matrix elements on that domain, not arbitrary subalgebra changes or an operator-norm Taylor expansion. The one-sided CHM charge used for a ball below is distinct from the full modular generator; their relevant local-unitary expectation differences agree, but the operators themselves do not.

Theorem 5.5 (Linearized gravity from entanglement, [39])**.** *In a holographic CFT dual to Einstein gravity in asymptotically AdS spacetime, the entanglement first law $\delta S = \delta\langle K_{\mathrm{mod}}\rangle$ for all boundary balls is equivalent to the linearized bulk Einstein equations $G^{(1)}_{\mu\nu} + \Lambda g^{(1)}_{\mu\nu} = 8\pi G_N\, T^{(1)}_{\mu\nu}$.*

The derivation uses the ball modular Hamiltonian [47]

$$K^B_{\mathrm{mod}} = 2\pi \int_B d^{d-1}x\, \frac{R^2 - |\mathbf{x} - \mathbf{x}_0|^2}{2R}\, T_{00}(x), \tag{4}$$

together with the Ryu–Takayanagi formula and the relation between bulk and boundary modular Hamiltonians [48]; complementary arguments are due to Jacobson [49].

Theorem 5.6 (Conditional holographic matching for an HSMI-generated family)**.** *Let $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ be a modular-stable candidate in a holographic CFT vacuum representation, and assume:*

(H1) the AdS/CFT correspondence holds;

(H2) the Ryu–Takayanagi formula and its quantum corrections hold;

(H3) the JLMS relation between bulk and boundary modular Hamiltonians holds;

(H4) the compared algebra family is an explicitly supplied HSMI/ball-region family whose generator P implements the boundary-region variation used in the holographic first-law argument;

(H5) the regulated entropy variation, or its relative-entropy replacement, satisfies the hypotheses of the Faulkner et al. theorem for all required balls.

Then Eq. (3) *describes the modular-generator variation within that family, and the holographic first law matches the linearized Einstein equations by the imported result:*

$$\begin{aligned} \textit{HSMI/ball family (H4)} &\Longrightarrow \delta K_{\mathrm{mod}} \xrightarrow{(H5)} \delta S = \delta\langle K_{\mathrm{mod}}\rangle, \\ \textit{(H1)–(H3)} &\Longrightarrow \delta(G_{\mu\nu} + \Lambda g_{\mu\nu}) = 8\pi G_N T^{(1)}_{\mu\nu}. \end{aligned} \tag{5}$$

The theorem does not show that a generic stability-selected algebra variation is an HSMI translation, nor that HAFF algebra flow is gravitational dynamics.

5.4 The circularity issue

Jacobson's thermodynamic derivation [34] presupposes a background geometry, while HAFF wants geometry to emerge from algebra. Three strategies are discussed in the literature. (A) In the algebraic programme, wedges are defined through modular inclusions without a background metric, and Theorem E.5 recovers a positive translation from an HSMI; constructing a sufficient net and showing that the candidates of this paper form such inclusions remain open tasks. (B) A bootstrap starts from a seed geometry, derives linearized equations from the first law, updates the geometry and iterates. (C) In holography the boundary theory supplies the algebra and the bulk geometry is derived from boundary data. Theorem 5.6 uses strategy (C) and assumes the HSMI/ball interface. A geometry-free reconstruction from the present candidate class remains open.

6 One source and a same-spacetime geometric test

The holographic matching above is not the only conditional route. Consciousness II [24] uses a free massless conformally coupled complex scalar in $D = d + 1 \geq 3$ dimensions. Its ball algebras, physical translation generator and vacuum are supplied in Minkowski spacetime. This matters: they are not the finite pointer algebra of a JT reader. What can be combined in this model is the prepared source, its modular charge and a geometric consistency test in that same spacetime.

In units $\hbar = c = 1$, the scalar curvature coupling is $-\xi\mathcal{R}\Phi^2/2$, with $\Phi^2 = \phi_1^2 + \phi_2^2$ and $\xi = (d-1)/(4d)$. Metric variation gives the conformally improved stress. For a ball $B = B(x_0, R, u)$, write

$$\begin{aligned} K_B &= 2\pi \int_B w_R T_{uu}\, d^d x, \qquad w_R = \frac{R^2 - |x - x_0|^2}{2R}, \\ H_B &= K_B + Q_B, \qquad Q_B = 2\pi\xi \int_{\partial B} {:}\Phi^2{:}\; dA. \end{aligned} \tag{6}$$

These are vacuum-subtracted expectation data with a fixed scalar boundary prescription. Surface readings mean controlled collar limits. H_B is the physical one-sided energy, K_B its bulk charge; neither is the full modular generator $-\log\Delta_B$. Keeping the same curvature coupling with Einstein coefficient $1/(16\pi G)$ gives Wald density $1/(4G) - 2\pi\xi\Phi^2$. The surface coefficient is therefore fixed by the same matter action, not chosen separately for every reader.

Proposition 6.1 (A common-source conditional bridge)**.** *Use the conformal scalar representation and surface prescription above. Let δT and h be smooth source and metric perturbations on an open domain. For the intrinsic geodesic ball of fixed proper radius set*

$$\delta F_B[h] = \delta\left(A(\partial B) - \frac{d-1}{R}V(B)\right), \qquad \eta_A > 0.$$

Then the following statements hold.

1. *The bulk condition*
$$\eta_A \delta F_B[h] + \delta\langle H_B\rangle - \delta\langle Q_B\rangle = 0 \tag{7}$$
for all centres, radii and rest frames is equivalent to $G^{(1)}[h] = (2\pi/\eta_A)\delta T$. It suffices to test the leading small-ball term at every centre and an open set of rest frames.

2. *At $\eta_A = 1/(4G)$ the same condition is $\mathcal{B}_B + \delta\langle H_B\rangle = 0$, where*
$$\mathcal{B}_B = \delta S_{\mathrm{W}}[\partial B] - \eta_A\frac{d-1}{R}\delta V(B) = \eta_A\delta F_B - \delta\langle Q_B\rangle$$
at linear order about the vacuum reference. The source normalization is $8\pi G$ in both forms.

3. *If selected ball subalgebras have vacuum-preserving normal conditional expectations and contain the bounded stress readings and the required surface readings, restricting one fixed physical state to them leaves Eq.* (7) *and its sourced Einstein tensor unchanged. A metric additionally requires common gravitational initial or boundary data and gauge identification.*

4. *Suppose a preparation has a smooth source with integrable energy density of strictly positive total energy, and that the source and response Fourier transforms belong to the tempered tensor-valued Radon measure class of Consciousness II. If consistency is tested at all centres and an open frame set, its nonzero-frequency data identify the canonical intrinsic response up to one normalization at each tested radius, within that paper's intrinsic, gauge-invariant, isotropic, compactly supported linear response class.*

Proof. The exact flat ball identity is $\delta F_B = -\int_B w_R G^{(1)}_{uu}$ [24, Theorem on the exact identity]. Subtracting Q_B from H_B in Eq. (6) gives the other integrand $2\pi\delta T_{uu}$. The leading coefficient of their sum is a nonzero multiple of $R^{d+1}(-\eta_A G^{(1)} + 2\pi\delta T)_{uu}$. An open frame set recovers the symmetric tensor; the field equation conversely makes every ball residual zero. The curvature derivative of $(1/(16\pi G) - \xi\Phi^2/2)\mathcal{R}$ gives the displayed Wald density, so its scalar part is exactly $-Q_B$ at this order. Conditional expectations preserve the vacuum modular restriction and the common stress readings, proving the third statement. For the fourth, a Radon Fourier measure supported only at zero would

give a constant source. Integrability forces its energy density to vanish, contrary to the positive energy. The nonzero-frequency identification theorem of Consciousness II now applies with its stated response hypotheses. These are consequences of that model's results, not an independent derivation of modular consistency. □

The preparation hypotheses are nonempty. In the same scalar theory, a local Weyl pulse gives a smooth improved stress with positive finite energy and a rapidly decreasing mass-shell convolution measure. A neutral pulse $U_\theta = \exp[i\theta(\phi_1(f)^2 + \phi_2(f)^2)/2]$ belongs to a proper symmetry-fixed stable hull when its bounded seed is permitted. It too has a smooth positive-energy nonzero-frequency source [24, Sections on prepared sources and selected algebras]. The permission to execute that pulse is an input; containing stress observables alone need not contain its generator.

There is a direct first-order comparison using this same preparation. Fix $\theta \neq 0$ and take the one-sided family $\psi_\epsilon = (1-\epsilon)\omega + \epsilon\omega_{U_\theta}$, $0 \leq \epsilon \leq 1$. Its source and collar tangents are exactly the neutral pulse's finite-amplitude differences. Choosing the constraint-compatible linear metric response to that conserved source makes Eq. (7) hold for every ball, with the same Wald coefficient. The mixture can be prepared by a recorded random choice of the permitted pulse or identity; that choice and the apparatus are charged. The prepared matter source is not the whole apparatus source when controls occupy the tested domain. This common-model comparison uses the Einstein coupling and gravitational constraint data as inputs, not outputs of the pulse.

A bounded non-scalar accessible observable with a permitted finite translation interval can instead be time-smoothed to give a bounded pulse of strictly positive finite energy. Local stress regularity and the Radon condition still require separate proof in that general route. Exact positive-energy translation permission on a nontrivial separated finite-dimensional algebra is impossible in the unique-vacuum setting; a finite clock is not that translation generator.

The fixed-source statement also survives finite adaptive reading within closed permissions. If $E : \mathcal{A} \to \mathcal{B}$ is a normal conditional expectation and every history-conditioned Heisenberg instrument preserves $\mathcal{B}$, initial states ρ and $\rho \circ E$ give the same record probabilities and, on positive-probability histories, conditional bounded source readings in $\mathcal{B}$ [24]. For each history, the pulled-back record effect and source observable stay in $\mathcal{B}$, where E is identity. Unbounded readings need absolute expectation convergence of spectral truncations. This permits real feedback that changes a common source; it forbids that same restricted feedback from recovering a discarded initial distinction. Enlarged controls or side information change the premise.

The bridge constrains the law rather than merely measuring one response: within its intrinsic response class, a nonzero spectral witness removes the higher-spatial-derivative ambiguity left by a homogeneous source. It does not select an entire metric from matter data; homogeneous gravitational solutions and common boundary data remain. Raw scalar H_B cannot replace the subtracted charge on all small balls: its surface term is $O(R^{d-1})$, while smooth bulk stress and geometry begin at $O(R^{d+1})$. Nor is the Wald coefficient match a modular first law for an arbitrary curved diamond. The area coefficient, region family, source reference, preparation permissions and Eq. (7) are still supplied.

6.1 Conservation before geometric interpretation

A source eligibility rule cannot remove unread stress at will. Q-RAIF NV [26, Section on physical selection and reaction] proves a limited obstruction: on a connected flat domain

of $D \geq 2$, a filter $T \mapsto a(x)T + b(x)\,\mathrm{tr}(T)g$ that conserves every smooth symmetric conserved T has constant a and $b = 0$. If it also kills every constant vacuum source, it is zero. This is a statement about that local algebraic filter family, not a no-go theorem for additional-field or nonlocal gravity models.

The same paper exhibits the appropriate alternative at the classical level. For an actual coupling field χ and canonical matter $\mathcal{L}_m = -\frac{1}{2}(\nabla\psi)^2 - U(\psi)$, let

$$S = \int \sqrt{-g}\left[-\tfrac{1}{2}(\nabla\chi)^2 - V(\chi) + f(\chi)\mathcal{L}_m\right], \qquad f > 0.$$

On both field equations,

$$\nabla^\mu(fT^m_{\mu\nu}) = \mathcal{L}_m\partial_\nu f = -\nabla^\mu T^\chi_{\mu\nu}.$$

The changed material coupling carries a compensating physical stress. This standard action-and-reaction identity is not a quantum total-source construction or a history-derived selector. It also does not cancel a vacuum term: shifting $\mathcal{L}_m$ by $-\rho$ changes both the total stress and the χ equation.

There is a concrete incompatibility to resolve before combining this mechanism with Proposition 6.1. A varying $f(\chi)$ changes the matter theory, so its ball modular Hamiltonian is not automatically the free conformal one. One may compare the prepared source in a causal domain where that interaction has ended and the specified conformal dynamics applies, with selector stress absent there or explicitly accounted for; or one must derive the interacting modular response anew. A separately prepared switching field plus the free CHM formula is not a common-model proof.

6.2 The second-order test

At the next order the source must follow the changed geometry. Consciousness II writes $g_\epsilon = \eta + \epsilon h + \epsilon^2 j + \cdots$ and $T_\epsilon = \epsilon t_1 + \epsilon^2 t_2 + \cdots$. Its corrected small-ball test yields

$$G^{(1)}[j] + G^{(2)}[h,h] = (2\pi/\eta_A)t_2$$

only after adding the slice term $-\frac{V_d}{2(d+2)}R^{d+1}[(\mathrm{tr}\,K)^2 - K_{ij}K^{ij}]$ to the area–volume functional, or choosing a geodesic slice with $K(p) = 0$. The source coefficient obeys the expanded covariant Ward identity, including its metric variation and changed propagation. A flat coherent pulse kept unchanged in the perturbed metric generally fails this identity. The same nonminimal action supplies the scalar propagation correction, the $\xi G_{\mu\nu}\Phi^2$ stress and the Wald coefficient.

This is a conditional leading two-jet stress law, not an exact finite-radius curved modular identity. An arbitrary bent slice already gives a false source without its extrinsic correction. The full quantum reference stress and preparation apparatus must also enter the same renormalized source. Thus a common action supplies necessary source and coefficient relations, while the actual curved modular response remains an additional problem.

7 The mechanism to be realized

A physical history must do two jobs at once: form and maintain the accessible structure, and supply the total source to which the geometric law responds. The finite return

of Section 9 establishes a source-to-operation edge inside an assigned gravity theory. Proposition 6.1 establishes what an all-ball consistency law requires in a different, higher-dimensional matter theory. Their relation is the following strong target.

Conjecture 7.1 (Common-action selection and geometric consistency)**.** *There is a local matter–controller theory in which readable histories form and maintain a nontrivial selected geometric structure, and whose causal effective action uses the same prepared fields, support and clock in its source and response. In a low-energy sector with a common cone, its selected ball data satisfy the geometric consistency condition of Consciousness II. Its first-order response is the corresponding Einstein response with one source normalization; its higher-order response includes the interaction, contact and clock work that produced the selection. Selection changes this response through physical couplings and sources, while a change of description at fixed matter history leaves the source unchanged. The sustainable family of Conjecture C.1 is a finite operating sector of this theory, with a response-to-operation gain that survives the growing-window limit.*

The conjecture has tests beyond assigning gravity to a massive apparatus. Removing the history's access to a command must remove the record-mediated component of the source, while holding the commanded interaction fixed and merely discarding a log must preserve it. Source identification must survive switched physical references; logical noise records alone fail this test in Proposition F.1. The same calibrated source must then pass the all-ball consistency test with fixed initial or boundary data. A law that reproduces only the JT dilaton or one binary return has not passed the higher-dimensional test. At second order the source, geometry and changed preparation must be varied together; entropy Hessians do not become SLD or spacetime metrics by name.

A model satisfying the conjecture must output a nontrivial selected sector from a family of retained histories; a prescribed sequence of couplings is not enough. With a supplied noise and readable seed, Foundation I gives a relative stable-hull construction. The next demand is that the same history dynamics prepares those inputs and that the resulting region data satisfy Eq. (7), with the total source and one area coefficient. The scalar surface match removes an independent matter boundary coefficient, but does not remove MC. This is the precise gap between common-action source formation and a selected gravitational law.

The hardest competing explanation is ordinary matter feedback on an unchanged gravitational stage. It explains every finite JT return here. To go beyond it, the selected localization family must predict a common geometric response without refitting the source reference, area coefficient or region prescription for each history. A permanent mismatch of that test would refute this common-action realization, even if its controller remained useful.

8 From a Selected Interface to a Physical Source

A calibrated interface acts through a field coupling whose outgoing stress sources the supplied JT theory. This gives a physical selection–source map; the origin of the gravitational law in Conjecture 4.2 is a separate question.

8.1 What records can and cannot do to a source

Let a classical record $R \sim \mathrm{Bernoulli}(p)$ control whether a physical operation prepares the matter state ψ or leaves the reference ω. Without reading R, the matter state is $\rho_p = (1-p)\omega + p\psi$ and its stress is the mixture of the two stresses. Deleting or ignoring R after the preparation does not act on the matter, so ρ_p and its source are unchanged. Changing the preparation probability p changes the source. The information side has a matching statement. For the joint state $\rho_{RM} = (1-p)|0\rangle\langle 0| \otimes \omega + p|1\rangle\langle 1| \otimes \psi$ and the reference $\pi_R \otimes \omega$, with π_R the marginal of R, additivity of relative entropy on direct sums and data processing give

$$D(\rho_{RM} \| \pi_R \otimes \omega) = p\, D(\psi\|\omega), \qquad D(\rho_p\|\omega) \le p\, D(\psi\|\omega) \tag{8}$$

[51]. Losing the record lowers the distinguishability available to a reader; it does not remove the matter source. A gravitational consequence of selection must therefore pass through a preparation or interaction, and the conditional geometry of a branch, the average of branch geometries and the geometry of the averaged source are three different objects.

8.2 The source of a calibrated feedback actuator

The companion paper [21] realizes a corrective flip of a degenerate bit as a field coupling. A right-moving free massless scalar with null coordinate u, current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$ is coupled through a real smooth switch g. We require a finite nonzero one-particle norm $\lambda^2 = \int_0^\infty \omega |\hat{g}(\omega)|^2 d\omega/(4\pi)$ and $\int (1+u^2)|g'(u)|^2 du < \infty$, where $\hat{g}(\omega) = \int g(u) e^{-i\omega u} du$ in the L^2 sense when needed. For example, $g \in H^1(\mathbb{R})$ together with the weighted derivative condition is sufficient. Smooth compact switches and the full-line packet below both belong to this class. The smeared current defines a Weyl unitary,

$$U = \exp\Big(-iX \int g(u) j(u)\, du\Big) = \exp\big[-i\lambda X (b + b^\dagger)\big], \tag{9}$$

where b is a normalized field mode with mean frequency $\bar{\omega}$ fixed by g, and $\lambda^2 \bar{\omega} = \frac{1}{4}\int g'^2 du$. With the drive mode in the coherent state $|\sqrt{N}\rangle$ and $2\lambda\sqrt{N} = \pi/2$, the coupling flips the bit with success probability $s_N = (1 + e^{-\pi^2/(8N)})/2$. From the current commutator $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$ one has $U^\dagger j(u) U = j(u) + \frac{1}{2} X g'(u)$ and

$$U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u) X j(u) + \tfrac{1}{4} g'(u)^2. \tag{10}$$

The cross term has zero expectation if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis; a Z-diagonal bit marginal alone does not suffice [21]. Writing $j = \psi(u) b + \psi^*(u) b^\dagger + j_\perp$, the incoming coherent beam has stress $T_{\mathrm{in}} = 4N[\mathrm{Re}\,\psi(u)]^2$ and each activation adds

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2. \tag{11}$$

This added stress is a property of the coupling. Its integral is the switching work $\lambda^2 \bar{\omega}$ supplied by the external switching agent.

The feedback task is that of the Imprint Chain [22] with the flip realized by this coupling. A hidden instrument label is estimated from calibration records with mismatch

probability q; each task step reads the bit with reliability ν, accepts feedback with probability h, switches g on when accepted and the reading is negative, and lets a bath with contraction c act. Each opportunity receives its own scheduled incoming beam, independent of the bit and earlier records. In the companion paper the stationary loss $L_\infty(q)$ and the stationary coupling rate satisfy [21]

$$p(q) = h\nu L_\infty(q) + \tfrac{h(1-\nu)}{2}. \tag{12}$$

Proposition 8.1 (Source–loss relation)**.** *At fixed drive, switch and task parameters, the stationary mean source per opportunity outside the apparatus is* $\langle T_{uu}\rangle_q = T_{\rm in} + p(q)F_g$. *For two calibrations,*

$$\langle T_{uu}(u)\rangle_q - \langle T_{uu}(u)\rangle_{q'} = h\nu\,[L_\infty(q) - L_\infty(q')]\,F_g(u). \tag{13}$$

Proof. The incoming beam is present at every opportunity. A coupling occurs with stationary probability $p(q)$ and, under the joint initial condition, adds F_g in expectation by Eq. (10). Equation (12) gives the difference. □

Better calibration lowers the task loss and, through it, the source. The coefficient comes from the same interaction that performs the correction; no separate rule maps a calibration error to an emitted pulse. The relation does not assign stress to knowledge. Deleting the calibration record after the commands have been executed leaves the outgoing source unchanged, in line with Section 8.1. The relation is a stationary average per opportunity; a single trajectory, with its fixed hidden match label, does not follow the population mean step by step.

8.3 A source set by learned quantum protection

The activation-rate mechanism is not the only way that retained records can change a source. Brain Anchor [57] constructs two degenerate qubits with a programmable coupling

$$A_a = Z_1 + \alpha a Z_2, \qquad U_a = e^{-iA_aQ_f}, \quad Q_f = \int f(u)j(u)du = \sqrt{k}(b + b^\dagger). \tag{14}$$

The real smooth compact switch f and positive gain interval are supplied. The gain α is unknown, and each encounter receives a fresh independent vacuum field. On $\mathrm{span}\{|01\rangle, |10\rangle\}$ the coupling is $(1 - \alpha a)Z_L$. Calibration records from the commands $\pm h$ have means $r_\pm = e^{-2k(1\mp\alpha h)^2}$, so $\log(r_+/r_-) = 8k\alpha h$. The finite-record certificate in that paper bounds $|1 - \alpha a|$, including command rounding and known drift. It does not promise an exact fixed algebra after finite training.

Proposition 8.2 (One residual controls protection and source)**.** *Fix the acquired command* a, *write* $m = 1 - \alpha a$ *and* $F_f = f'^2/4$, *and keep the initial field independent of the logical body and retained ancillas. For every logical input, one encounter has added mean stress* m^2F_f *and switching work* m^2E_f, *with* $E_f = \int F_f du$. *After* M *independent identical encounters, with the output ports discarded and no interleaved gates or active recovery, its passive protection error is*

$$\epsilon_M = \tfrac{1}{2}(1 - e^{-2Mkm^2}). \tag{15}$$

For a single pulse with stress moments $E_j = \int u^j F_f du$ and the initial data of Proposition 8.4, the late invariant is

$$\mathcal{C}(m) = H_0^2 + 2\kappa H_0 m^2 E_1 + \kappa^2 m^4 (E_1^2 - E_0 E_2). \tag{16}$$

Thus retained history can change the physical protection and the gravitational response at fixed pulse shape and activation schedule.

Proof. The current shift is $j \mapsto j + A_a f'/2$. The centered independent field input makes the stress cross term vanish for every logical state, and $A_a^2 = m^2 I_L$ on the code. Its characteristic function multiplies the logical off-diagonal by e^{-2km^2} per encounter; a dephasing channel's half-diamond error is its phase-flip probability. Time ordering may give a factor $e^{i\vartheta_f A_a^2}$, which is scalar on the code but not on the whole two-qubit register. Finally substitute $m^2 F_f$ in the fixed-initial-data source solution below. □

This link does not require a rule that emits a pulse only after a wrong answer. The same interaction produces both the residual error and its source, even when the logical body is entangled with retained memory. At $m = 0$ the body decouples in the code; a noiseless stored qubit need not continuously export entropy. The finite-rate entropy-export bound of Section 10 applies to actual renewal, not to every protected state. Calibration itself adds field energy $2N(1 + \alpha^2 h^2)E_f$ for N independent tests at each sign. Preparations, measurement, the controller and switching mechanism add further charges. The source suppression during operation alone is not a total-energy saving.

Brain Anchor also treats an unknown centered Gaussian variance K_ω. Logical records at $0, \pm h$ give $K_\omega = -\log r(0)/2$ and $\alpha = [\log r(h) - \log r(-h)]/(8hK_\omega)$; two settings suffice when the signs of $1 \mp \alpha h$ are known. Its finite-record mismatch certificate needs no supplied temperature label. The coherence is now $e^{-2K_\omega m^2}$, but the coupling-added source remains $m^2 F_f$ on centered independent input.

Incoming stress must still be added. A u-clock chiral KMS input has background $\pi/(12\beta^2)$ in natural units. Within a supplied bounded one-temperature family, strictly decreasing K_β permits a finite-error background estimate; a general Gaussian variance does not determine incoming energy. At common preparations and initial data, the linear dilaton equation compares coupling-source differences. Equation (16) cannot be used for the full source after dropping its thermal background, nor is u thermality the rapidity-clock reference of Section 10.

The same command can also operate on one reused vacuum field. For nonnegative compact switches f_j on disjoint intervals, with no noncommuting logical gates, the smeared currents commute but have negative correlations. Brain Anchor gives exact coherence $e^{-2m^2 k_M}$ with $k_M = \langle (\sum_j Q_{f_j})^2 \rangle \le Mk$, and added source $m^2 \sum_j f_j'^2/4$ of energy $Mm^2 E_f$. Its JT response uses the actual pulse-train moments and initial coefficients in a valid operating domain. No independent-port approximation, thermal input or infinite-lifetime worldline is inferred.

There is a finite causal pulse in this construction. For example, set $f(u) = A \exp[-1/(1 - (u-3)^2)]$ on $2 < u < 4$, zero outside, and normalize its one-particle norm to $k = .2$. Numerical quadrature gives $(E_0, E_1, E_2) = (.4568522, 1.3705565, 4.3215400)$. For $H_0 = 1$, $\kappa = .05$ and $\alpha = 1.1$, the supplied command $a = 48/55$ has residual $m = .04$ and gives $\mathcal{C} \simeq 1.00021929$; the comparison $a = 1$ has $m = -.1$ and $\mathcal{C} \simeq 1.00137053$. These are deterministic evaluations of the displayed pulse, with code and outputs in `supplemental/pulse_debit`, not statistics of a calibration pilot. For a random learned

command, the invariant of the mean source uses $(\mathbb{E}m^2)^2$, whereas the mean branch invariant uses $\mathbb{E}m^4$ in the quadratic term. The compact source ends before a later boundary echo is scheduled, but the calibrated wall and timing instrument remain the inputs of Section G.1.

Stress readback also has an exact blind direction. At fixed $a_0 > 0$, the gains $\alpha_\pm = (1 \pm d)/a_0$ give residuals $\mp d$ and therefore the same protection channel and source. Field parity conjugates their full code interactions while leaving the vacuum and stress invariant. No parity-covariant stress instrument with invariant outcome labels can distinguish them, nor can their mean JT responses at common initial data. Their correcting commands $1/\alpha_\pm$ differ. Probes at $a_0 \pm h$ break this ambiguity through the opposite signs of $8k\alpha h(1-\alpha a_0)$ in their log contrast. A gravitational response is therefore not a complete calibration reference for the interface that produced it. Off-frame coupling probes, or a signed field quadrature, are additional access, with their own sources and costs. Full output-field parity and a conditional logical pulse can instead restore the stored state without identifying the gain, as shown in Ref. [57]. This recovery leaves the emitted source unchanged. Active state correction and learning a match that suppresses the actual source are therefore distinct operations.

The gain is a physical actuator parameter, not a freely changed description. The selected finite code is not the geometric algebra of Section 5, and the supplied JT action still supplies the gravitational response. A link to continuous modular selection requires that further interface; it is not obtained by identifying a protected M_2 sector with a half-sided modular inclusion.

8.4 A physical seed changes the source

The rate-controlled model changes how often a fixed interaction occurs. The coupled observable can itself change, giving a direct relation between an exactly represented algebra and a source. Keep the incoming field, switch and activation schedule fixed. On a qubit put $P_\theta = (I + \cos\theta\, Z + \sin\theta\, X)/2$ and use $U_\theta = \exp\big[-i\lambda P_\theta(b + b^\dagger)\big]$. The projection selects a pointer algebra and its source population under this fixed schedule.

Proposition 8.3 (Selected projection and outgoing stress)**.** *For $\lambda \neq 0$, an initially independent qubit state ρ and normal field state σ, the exactly represented algebra of the induced qubit channel is $\{P_\theta\}'$. Let $r_\theta = \mathrm{tr}(\rho P_\theta)$. The stress formula below requires its displayed current moments to exist. The outgoing field is*

$$(1 - r_\theta)\sigma + r_\theta e^{-i\lambda Q}\sigma e^{i\lambda Q}, \qquad Q = b + b^\dagger,$$

and its stress is

$$T_\theta(u) = T_{\mathrm{in}}(u) + r_\theta \left[g'(u)\langle j(u)\rangle_\sigma + \tfrac{1}{4}g'(u)^2\right]. \tag{17}$$

Thus two selected projections can change the source at fixed global input state and fixed activation schedule.

Proof. The quadrature has non-atomic spectral distribution in a normal state, so the multiplicative-domain theorem of Q-RAIF C gives $\{P_\theta\}'$ [14]. Expanding U_θ in its two projection blocks and tracing the qubit gives the field mixture; off-diagonal blocks have zero trace even when ρ does not commute with P_θ. In the current identity replace the Pauli X by P_θ and use $P_\theta^2 = P_\theta$. Initial independence gives (17). □

For $\rho = (I + 0.8Z)/2$, the seeds P_0 and $P_{\pi/2}$ select different pointer algebras, with $r_0 = 0.9$ and $r_{\pi/2} = 0.5$. In the worked coherent packet, the cross profile is $8\lambda\sqrt{N}\,\mathrm{Re}\,\psi\,\mathrm{Im}\,\psi$. Its first three moments vanish, so the invariant increment is (70) with p replaced by r_θ. With the table's $N = 16$ and background it is 2.0807581×10^{-5} versus 1.1559770×10^{-5}. The full source also contains the cross profile, which enters the local dilaton even though its first three moments do not enter the final invariant.

Here the source weight is derived from the state's population in the selected projection; it is not assigned to an external mismatch label. Changing the projection means changing a physical interaction, not renaming an unchanged algebra. The two algebras are abstractly isomorphic but are differently embedded relative to the fixed state and coupling. This supplies a seed–interface–source link. It does not identify that finite-dimensional pointer algebra with a Type III geometric candidate or explain why the coupling selects this seed rather than another. Those are the remaining selection and geometry bridges.

8.5 Response of a given Jackiw–Teitelboim model

Take the JT action [52, 53]

$$I = \frac{1}{16\pi G}\int \sqrt{-g}\,\Phi(R + 2/L^2) + I_{\mathrm{matter}}, \qquad ds^2 = -\frac{4L^2\,du\,dv}{(1+uv)^2}, \tag{18}$$

with the usual boundary terms, $\kappa = 8\pi G$, $D = 1 + uv$ and the vacuum dilaton $\bar\Phi = H_0(1 - uv)/D$, $H_0 > 0$. The field equations are $\nabla_\mu\nabla_\nu\Phi - g_{\mu\nu}\Box\Phi + g_{\mu\nu}\Phi/L^2 = -\kappa T_{\mu\nu}$, and in these coordinates

$$\Phi_{uu} + \tfrac{2v}{D}\Phi_u = -\kappa T_{uu}, \qquad \Phi_{vv} + \tfrac{2u}{D}\Phi_v = -\kappa T_{vv}, \qquad \Phi_{uv} + \tfrac{2}{D^2}\Phi = \kappa T_{uv}. \tag{19}$$

The matter is a conformal theory with matched left and right central charges, and the state-independent anomaly is removed by subtracting the same background. The left-moving input is unchanged, and the state-dependent source is right-moving, $T_{uu} = F(u)$ with $T_{uv} = T_{vv} = 0$.

Proposition 8.4 (Response with fixed initial data)**.** *Let F be integrable with $\int |u|^2|F|du < \infty$. On a connected coordinate domain with $D \neq 0$, write the perturbation as $\Phi - \bar\Phi = (A(u) + vB(u))/D$ and impose the initial coefficient data $A(u), B(u) \to 0$ as $u \to -\infty$. For a source bounded below in support this is equivalent to zero perturbation before the source. The unique solution with these data is*

$$\Phi = \bar\Phi + \frac{A(u) + vB(u)}{D}, \qquad A = -\kappa\int_{-\infty}^{u}(u-z)F(z)dz, \qquad B = -\kappa\int_{-\infty}^{u} z(u-z)F(z)dz. \tag{20}$$

The perturbation is linear in F. With $E_j = \int u^j F\,du$, after the source (or in the asymptotic limit for sources with tails) the invariant $\mathcal{C} = \Phi^2 - L^2(\nabla\Phi)^2$ takes the vacuum value

$$\mathcal{C} = H_0^2 + 2\kappa H_0 E_1 + \kappa^2(E_1^2 - E_0E_2), \tag{21}$$

and inside the support $\partial_u\mathcal{C} = \kappa F\,(2H_0u + uA - B)$.

Proof. Write $\psi = D\,(\Phi - \bar\Phi)$. The vv equation of (19) reads $\psi_{vv} = 0$, so $\psi = A(u) + vB(u)$, and the other two equations reduce to $A'' = -\kappa F$, $B'' = uA''$ and $B' = uA' - A$, which

Eq. (20) satisfies. The absolute first and second moments make both retarded coefficients tend to zero at $-\infty$. Two solutions differ by $\alpha + \beta u + \gamma v - \alpha uv$, whose coefficients are $A_h = \alpha + \beta u$ and $B_h = \gamma - \alpha u$. Their two zero limits force $\alpha = \beta = \gamma = 0$. Every vacuum solution has numerator $\alpha + \beta u + \gamma v - \alpha uv$ and invariant $\alpha^2 + \beta\gamma$; after the source $\alpha = H_0 + \kappa E_1$, $\beta = -\kappa E_0$, $\gamma = \kappa E_2$. The balance law follows by differentiation. □

The coefficient condition cannot be replaced by $\Phi - \bar{\Phi} \to 0$ at each fixed v. The homogeneous solution $\eta v/(1 + uv)$ has that limit for every fixed v along nonsingular points. It even tends to zero uniformly on the subdomain $u < 0$, $v \leq 0$, since its absolute value is at most $|\eta|/|u|$; no uniform limit near $D = 0$ is implied. Adding it leaves the source unchanged but changes the late invariant by $-\kappa E_0 \eta$. It is excluded by $B(-\infty) = 0$. Source comparisons below hold H_0, the coordinate matching and both coefficient boundary data fixed.

The JT curvature stays $-2/L^2$; the dilaton and the invariant respond. A dilaton difference is not by itself a higher-dimensional curvature or a mass, and comparing boundary proper times needs a boundary normalization.

The matching condition matters. Set $A_F, B_F \to 0$ at $+\infty$, or exactly to zero after a compact source, instead of the initial data of Proposition 8.4. The final-state-matched solution differs from the retarded solution by

$$A_F = A_R + \kappa(uE_0 - E_1), \qquad B_F = B_R + \kappa(uE_1 - E_2).$$

For the chiral diffeomorphism states of Consciousness I [25], with a Cut at a to the left of the support, the relative entropy is $D_a = 2\pi \int (u - a)F(u)\, du$. The final-state-matched solution gives $\delta\Phi(a, 0)/(4G) = -D_a$, while the initial-state-matched solution gives $\delta\Phi(a, 0) = 0$ at a Cut before the pulse. The first identity compares an earlier section with a matched final state; it does not say that a pulse removes area before it arrives. The dilaton–modular-energy relation itself is known [58]; what is used here is its type-correct pairing with a specified source and matching condition.

The same equations show that a null-only record of the source is incomplete. A genuine vacuum-density shift $T_{\mu\nu} = -\delta\rho\, g_{\mu\nu}$ has zero null components but gives the particular solution $\delta\Phi = \kappa L^2 \delta\rho$. A probe that reads only null energy cannot certify that a source is absent, which is the source-completeness condition relevant to the vacuum curvature test of Ref. [26].

9 Response that changes the next operation

A source-to-geometry calculation becomes a return only when the geometry changes an actual record and that record changes a later interaction. The readout permission matters. In the minimally coupled JT sector, instruments on the same matter metric with the same inputs have identical adaptive record laws even when the dilaton differs (Proposition G.1). A boundary tied to a fixed dilaton value evades that obstruction. Its echoes reconstruct the three vacuum coefficients, and a source-compensated inverse includes finite probe backreaction (Propositions G.2 and G.3). An exact current contact instead obtains an invariant-dependent bit, including the contact's own source (Propositions G.5 and G.6). The bit selects the next physical projection. Its same-field conditional covariance remains part of the history; detector replacement does not reset the field.

These are finite physical records, not permission to read an exact metric. The worked current contact has total variation only 2.3650×10^{-4} between its two initial geometries. Its fixed-cutoff energy budget also forbids indefinitely many contacts of unchanged positive source charge. Appendix G contains the timing, stress and accumulated-source bounds.

9.1 An acquired record drives a new material source

A record can act without first being converted into a supplied mechanical initial condition. Keep the two classical scalar fields and the material action of Section J.4, and append $i\langle\Psi,\dot\Psi\rangle - \langle\Psi, H_c\Psi\rangle$ to its proper-time Lagrangian. The normalized state Ψ contains a finite history clock, probe P, record R, replacement blank B and an old purification Q. Set

$$H_c(\Phi,y) = H_{\rm hist} + E_{\rm off}\mathbf{1} + \lambda(\Phi-\phi_0)Z_P + \frac{\chi_R}{2}P_R(y-d)^2, \qquad P_R = |1\rangle\langle 1|_R, \quad \chi_R \geq 0. \tag{22}$$

$H_{\rm hist}$ is the positive gate-conjugated clock of Q-RAIF C. Its first nonidentity edge performs the Y_P acquisition C_Y, the next performs a true SWAP of P and B, and the remaining edges are identities. There is no proper-time switch. Initially $R = 0$, so the spring has zero expectation. The actual acquisition opens its force channel. $H_c \geq 0$ while $E_{\rm off} > |\lambda(\Phi - \phi_0)|$. The target d, couplings and initial clock are prepared hardware. The scalar fields, mechanical coordinates and geometry use an Ehrenfest mean-action approximation [61, Section 2]; the finite quantum core and its old reference are retained in full, without sampling or collapsing R.

Here θ is a material angle, y its storage coordinate, $I, J, k, K > 0$ their inertia and elastic parameters, and $U(\Phi)$ the material support potential. Proper-time derivatives obey $I\ddot\theta + k(\theta - y) = -\gamma\dot\theta$, $\gamma = g_+^2 + g_-^2$, with no incoming classical waves. Their emitted powers are $P_\pm = g_\pm^2\dot\theta^2$. The mechanical stored energy is $e = I\dot\theta^2/2 + J\dot y^2/2 + k(\theta-y)^2/2 + Ky^2/2$, and $w = \dot\Phi$ along the wall. The positive wall junction and its two invariant charges are specified by Eqs. (128) and (137); all of these mechanical and gravitational data are supplied.

Let $p = \langle P_R\rangle$, $z = \langle Z_P\rangle$ and $E_c = \langle H_c\rangle$. Variation of this one action gives

$$\begin{aligned} i\dot\Psi = H_c\Psi, \qquad F_y = \chi_R(y-d)p, \qquad F_\Phi = \lambda z, \\ J\ddot y + k(y-\theta) + Ky = -F_y, \qquad \dot e = -P_+ - P_- - F_y\dot y, \qquad \dot E_c = F_y\dot y + F_\Phi w. \end{aligned} \tag{23}$$

The angle equation and $P_\pm = g_\pm^2\dot\theta^2$ are unchanged. Here p is an expectation obtained by variation, not a decoded number fed to a separate control policy. The mass and its tangent balance are

$$M = m_{\rm rest} + U + e + E_c, \qquad \dot M = (U' + F_\Phi)w - P_+ - P_-. \tag{24}$$

Consequently Eq. (137) holds with U' replaced by $U' + F_\Phi$ in its acceleration equation; both invariant and null-clock equations retain their directions. Differentiating the mass junction with these equations gives zero. The separate work balances are

$$\begin{aligned} e(\tau) + Q_{\rm out}(\tau) + \int_0^\tau F_y\dot y\, ds = e(0), \\ E_c(\tau) - E_c(0) = \int_0^\tau (F_y\dot y + F_\Phi w)\, ds. \end{aligned} \tag{25}$$

The clock interaction and conditional spring are included in E_c. The spring breaks the symmetry of Proposition J.6; F_Φ cannot be set to zero in this coupled calculation.

Proposition 9.1 (A nonzero record-mediated source return)**.** *Take two initial geometries with the same $\Phi = \phi_0$, $w = 0$, mass, mechanical data and quantum input, but different $\mathcal{C}_-$; set $\mathcal{C}_+$ by the mass junction. In the positive exterior domain, either $\lambda = 0$ or $\chi_R = 0$ makes their proper emitted powers identical. With both interactions present, there are positive finite-clock data for which the emitted powers differ on a nonempty finite interval. Their record, mechanical response, source and geometry obey the same action and work account.*

Proof. For $\chi_R = 0$, the mechanical subsystem is independent of the reader and geometry. For $\lambda = 0$, the joint mechanical and quantum equations are independent of geometry. Either condition fixes identical mechanical sources for the two initial geometries.

For a nonzero example choose $\theta(0) = \dot\theta(0) = 0$, $y_0(y_0 - d) \neq 0$ and probe state $(I + \upsilon X_P)/2$, $\upsilon \neq 0$, with a blank R and its old purification held. Let $r \geq 1$ be the C_Y edge, $a_j > 0$ the clock edge strengths and $G_r = \prod_{j=1}^{r} a_j$. The initial acceleration difference δa follows from the junction equations, so that $\delta\Phi(t) = \delta a\, t^2/2 + O(t^3)$. The shortest-path Dyson expansion gives

$$\delta p(t) = K_r t^{2r+3} + O(t^{2r+4}), \qquad K_r = \frac{\upsilon G_r^2 \lambda \delta a}{6r!(r-1)!(r+3)}. \tag{26}$$

An insertion of $\lambda\delta a\, t^2 Z_P/2$ after $j < r$ forward edges contributes the time integral $t^{r+3}(j+1)(j+2)/(r+3)!$. Their sum is $t^{r+3}r/[3(r+3)r!]$. Using $C_Y^* P_R C_Y = (I + Y_P)/2$ in the interference with the unperturbed shortest path gives the coefficient above. The insertion after all r edges has Hermitian $P_R Z_P$ in that interference and zero real part. Further graph paths and spring insertions raise the order; the spring annihilates the blank record before acquisition.

Integrating the mechanical equations then yields

$$\delta(P_+ + P_-)(t) = -\frac{2\gamma k^2 y_0 \chi_R (y_0 - d) K_r}{I^2 J (2r+4)(2r+5)(2r+6)} t^{2r+7} + O(t^{2r+8}). \tag{27}$$

The baseline angle velocity is $k y_0 t/I + O(t^2)$. Different initial $\mathcal{C}_-$ give a nonzero δa for generic supported data, since initially $\dot w = -U'\beta_+\beta_-/M - \phi_0/L^2$. The displayed coefficient is nonzero for the stated nonzero parameters. The coupled finite-dimensional equations are analytic on their strict positive exterior domain. Local existence and continuity therefore give a nonempty admissible family, not just a numerically resolved difference. □

The proof isolates the causal mechanism analytically, rather than attributing a small numerical difference to geometry. Blocking either sensing or record actuation removes the source contrast. Keeping the record but blocking actuation preserves a readout difference while removing the later source difference. Its numerical realization and complete mass/work checks are in Appendix J. The two histories here differ in initial gravitational data; they do not purport to generate those data from algebra selection.

For histories with common initial geometric constraints, the distinct spatial construction of Proposition J.7 has a nonzero source–dilaton–sensor path and a later material response. Its finite-width regulator, support and synchronization are physically varied.

The complete material response also contains ordinary clock and force effects; only the separately isolated path is gravitational. A third construction, Proposition J.8, couples the actual full-mode quartet state to paid clock/director fields and constrained JT bulk evolution. It supplies a local Ward identity and a causal prepared bulk domain, not moving-wall gate matching or the full thermal apparatus.

A finite record error alone does not control an unbounded stress. The response comparison uses weighted source bounds (Proposition A.1), and the coupled trajectory–reader bound requires domain-preserving strong solutions and the stated bounded-coupling derivative assumptions (Proposition A.2). Appendix A gives these estimates and the separate full-mode clock-work matching result. None converts fixed-window existence into sustained operation.

10 Maintenance and its gravitational price

10.1 Entropy removal and the source it can constrain

A fixed unitary correction can return its control port coherently. Removing entropy injected by unobserved noise is a different task. To price it, the state whose entropy is removed must include the controller's usable side information. The heat account must also distinguish a thermal reservoir from a supply of pure memories.

HAFF G supplies a record-driven program and executor in which selecting a physical coupling changes its source under one fixed executor [55]. The dephased quartet below adds a specified full renewal debit. It does not identify the two material realizations or inherit a thermal controller from that program loop.

Proposition 10.1 (Maintenance entropy account)**.** *Let $V = SC$ include the maintained system and every retained controller and side-information register. A correction acts unitarily on finite-dimensional VAB, initially in $\rho_V \otimes \alpha_A \otimes \tau_B$, where $\tau_B = e^{-\beta H_B}/Z$, $\beta = (k_B T)^{-1} > 0$. Put*

$$s = S(V) - S(V'), \qquad p = S(A') - S(A), \qquad Q = \operatorname{tr} H_B(\rho'_B - \tau_B).$$

Then, with $I_{V:A:B} = S(V) + S(A) + S(B) - S(VAB)$,

$$\beta Q + p = s + I_{V':A':B'} + D(\rho'_B \| \tau_B) \ \geq s. \tag{28}$$

If the marginal entropy of C is returned, s equals the decrease of $S(S|C)$. In a sequence of such corrections, a bounded total purity consumption and a positive asymptotic entropy-removal rate $\dot{s}$ imply $\liminf Q_{[0,t]}/t \geq k_B T \dot{s}$.

Proof. Unitary invariance of total entropy and the initial product give $\Delta S(V) + \Delta S(A) + \Delta S(B) = I_{V':A':B'}$. Also $\beta Q = \Delta S(B) + D(\rho'_B \| \tau_B)$. Combine the identities and sum over corrections. The rate statement follows when the accumulated p divided by elapsed time tends to zero. □

This is the Landauer entropy balance, with the memory-purity term displayed explicitly; the equality and the role of accessible side information are standard [13, Theorem 3 and Section 5.1]. For initially correlated or nonthermal inputs the same calculation instead gives

$$\beta Q + p = s + (I_{\text{out}} - I_{\text{in}}) + D(\rho'_B \| \tau_B) - D(\rho_B \| \tau_B).$$

These consumed correlations and bath free energy can pay for correction. A classical copy of the error permits a reversible conditional flip with $s = 0$, although the system's marginal entropy decreases. Negative quantum conditional entropy gives another reason to include the side information. A pure-memory swap has $p = s$ and can have $Q = 0$. A retained oscillator's closed unitary loop also has no entropy removal by itself. Thus a count of corrections or a marginal noise entropy is not automatically $\dot{s}$. In a cyclic device $\dot{s}$ must be established from its actual noise and controller dynamics. For nondegenerate systems, heat and supplied work are further separated by the change in internal energy.

Proposition 10.2 (A finite-rate chiral exhaust floor)**.** *Consider a unitary, zero-chemical-potential chiral conformal exhaust with central charge $c_{\mathrm{ex}} > 0$, thermal input at $T \geq 0$, and an extensive entropy-current account. Energy and temperature are defined by the same translation generator and clock. Assume that the outgoing modes export net dimensionless entropy current $x \geq \dot{s} > 0$, where $\dot{s}$ is the actual maintenance entropy-removal rate after all retained side information and sublinear resource consumption are accounted for. The entropy-current maximum at fixed energy is the thermal one. Then the excess heat current over the incoming thermal stream obeys*

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2. \tag{29}$$

Proof. Write $a = \pi c_{\mathrm{ex}} k_B^2/(12\hbar)$. The thermal energy and dimensionless entropy currents are $J_E(T) = aT^2$ and $J_S(T) = 2aT/k_B$. The first is the standard chiral CFT current [11, Eq. (1)]; the second follows from its extensive Gibbs entropy. Entropy maximization at fixed energy is the Gibbs variational principle, underlying the single-channel information-flow bound [10]. Choose $T_* = T + k_B x/(2a)$ so that the thermal output has the required entropy. Any output with at least that entropy has energy current at least aT_*^2. Subtracting aT^2 gives

$$\dot{Q} \geq k_B T x + \frac{k_B^2 x^2}{4a} = k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}.$$

The expression increases for $x \geq 0$. □

The current account is a physical condition on the exported modes, not the entropy of a Type III local interval. Equation (44) establishes the needed joint entropy rate in a fresh-port account; matching those ports to a conformal current must preserve its capacity and entropy budget. At fixed capacity and positive rate the quadratic term survives even at $T = 0$. Thermal output saturates the channel inequality, but this alone does not construct a correcting apparatus. The currents are long-window, extensive Gibbs limits. For an observation window τ with net exported entropy X_τ, bounded rates and sublinear boundary corrections give

$$Q_\tau \geq k_B T X_\tau + \frac{3\hbar X_\tau^2}{\pi c_{\mathrm{ex}} \tau} + o(\tau).$$

The current formula is not an exact inequality for an arbitrarily short finite packet. Applying the Gibbs bound to the entire output also includes correlations between packets; it does not require each output to be a separate thermal state.

10.2 A conditional gravitational reading

The price becomes a geometric reading only with a physical transport map. The following JT coefficient reads the total routed heat in one matched clock and reference. It does not assign curvature to entropy that remains in a memory or exits through another channel.

Proposition 10.3 (A heat account read through the dilaton)**.** *Compare two JT solutions with the same H_0, boundary coefficient data and coordinate and clock normalization. Suppose their source difference F is the exhaust of the correction in Proposition 10.1, with finite absolute second moment and $\int F du = Q$. This last equality requires that the specified outgoing channel carry all the heat in the same energy normalization. Then its retarded coefficient difference obeys*

$$-\frac{A'(+\infty)}{\kappa} = Q \;\geq\; k_B T(s-p). \tag{30}$$

For a compact exhaust and any $u_2 > u_1$ after it, the left side is exactly $-[A(u_2) - A(u_1)]/[\kappa(u_2-u_1)]$. No pointwise sign of F is required.

Proof. Differentiate Eq. (20): $A'(u) = -\kappa \int_{-\infty}^{u} F(z)dz$. Take the late limit and apply Eq. (28). □

The reference for Q is the incoming thermal reservoir transported through the same exhaust, before its heat uptake; H_B is its u-translation generator. A thermal background has a nonintegrable stationary flux, so the JT solution here is the difference from that same transported thermal background. The identification $E_0 = Q$ is an energy-transport hypothesis: other exhaust channels, energy stored in the device, work beams crossing the observation surface and redshift must be accounted for separately. With a fixed transport factor $E_0 = \zeta Q$, Eq. (30) acquires the same factor ζ. The coefficient slope is read from the dilaton in the declared matched coordinate frame, not an invariant under arbitrary changes of boundary clock. The relation links an actual entropy-removal process to a specified dilaton reading. The JT dynamics and heat-routing map are its geometric inputs.

For a matched running single-chiral source, the marginal-horizon balance is $dS_h/d\eta = 2\pi F\xi^2$. If the charged heat is exactly $F\xi^2$ and the physical input is thermal at $T_H = 1/(2\pi)$ in the same rapidity clock, the long-window bound becomes $\overline{\dot{S}_h} \geq \dot{s} + 6\dot{s}^2/c_{\rm ex}$. The actual thermal-source and reference-clock terms, positive-domain conditions and nonthermal contact excess are retained in Appendices B and H. A reparametrized vacuum is not a prepared heat reservoir.

11 Selection and response over growing windows

11.1 A conjecture on a sustainable selection–response loop

The unresolved question is whether one reused local body can combine selection, renewal, sharp exhaust and informative geometric feedback. Fix two physical preparations on a logical receiver A and its current purifier R:

$$|\Phi_Z\rangle = \sqrt{1-\epsilon}|00\rangle + \sqrt{\epsilon}|11\rangle, \qquad |\Phi_X\rangle = (H_A \otimes I_R)|\Phi_Z\rangle, \quad 0 < \epsilon < 1/2.$$

The logical labels denote Q-RAIF C's neutral pair codes, with their spectator sea retained. Here R is the paid, resettable current purifier in the third and fourth body species, available to joint neutral gates. Old references outside this source quartet are denoted R_{old} and are not acted upon. The quartet is a resettable working source alongside the series' protected unknown content H, not a replacement for it; access to the current R comes from those paid body species and joint neutral gates, not from an old-purifier permission. Each cycle's inaccessible environment applies the same pointer-aligned channel

$$\mathcal{D}_q(\rho) = (1-q)\rho + qZ_A\rho Z_A, \qquad 0 < q < 1/2, \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

No noise error label is available to the apparatus. Renewal returns the complete quartet, including the current purifier.

Proposition 11.1 (Pointer alignment and the complete renewal debit)**.** *For $\rho_p = \mathcal{D}_q(|\Phi_p\rangle\langle\Phi_p|)$, return to its specified pure quartet removes entropy $h_p = S(\rho_p)$, where*

$$h_Z = h\left(\frac{1-\sqrt{1-16q(1-q)\epsilon(1-\epsilon)}}{2}\right) > 0, \qquad h_X = h(q) > h_Z. \tag{31}$$

The receiver-only debits are 0 for Z and $h(\epsilon + q(1-2\epsilon)) - h(\epsilon)$ for X. Approximate quartet return subtracts its residual joint entropy from h_p, with finite-dimensional entropy continuity controlling the error. Program updates, sensors and usable controller registers enter the full maintenance debit in Proposition 10.1 as well.

Proof. The mixture of $|\Phi_p\rangle$ and $Z_A|\Phi_p\rangle$ has nonzero eigenvalues $(1 \pm d_p)/2$, with $d_p^2 = 1 - 4q(1-q)(1 - |\langle\Phi_p|Z_A|\Phi_p\rangle|^2)$. The overlaps are $1-2\epsilon$ and zero. Strict monotonicity of h on $(0, 1/2)$ proves the contrast. Tracing R gives the receiver debits. The reference marginal is unchanged by the noise, so the difference between each complete and marginal debit equals the receiver–purifier mutual information lost to the noise. □

Pointer alignment lowers the complete renewal price without making it zero. For $\epsilon = .1, q = .05$, $h_Z = .0877516288$ and $h_X = .1985152433$ nats per cycle. The marginal Z state is unchanged, but its purifier correlations are not. The loop must establish that its additional charged operations preserve an actual complete-debit contrast $\bar{s}_X > \bar{s}_Z$.

11.2 Three distinct targets

The finite return is target A: an actual acquisition changes a source, its response changes another acquisition, and the retained history remains in the same finite mean-action evolution. Target B is sustained thermal renewal of the body, controls and ports over growing runs while the independent unknown content stays held. Target C combines that renewal with a sharp complete exhaust and nonvanishing causal response gain. A does not prove B, and B alone does not prove C.

Conjecture C.1, stated in full in Appendix C, makes C quantitative. At fixed cycle rate and exhaust capacity, it requires complete-output sharpness, normalized whole-window response contrast, and a positive average gain at *all* declared decision opportunities on the actual-run history measure. Every sensor, program update, delay, buffer, work bank and current purifier enters the account; old inaccessible references are not reset. Resource size may grow with the run, while uncharged purity and correlation consumption have no positive asymptotic rate.

This target has real obstructions. Accurate direct purifier dumping retains cross-species redundancy and cannot make the complete Gibbs deficit vanish. Individually sharp program phases still pay a positive Jensen variance cost when their entropy rates differ at nonzero duty fractions. An exploration scheme whose only response updates occur on a vanishing fraction of opportunities has vanishing average gain, even with infinitely many such updates. These facts do not exclude every joint protocol; they exclude tempting combinations of successful components. The complete statements and resource clocks remain in Appendix C.

The following fixed-source converter supports one part of the target. It does not supply the adaptive source's product structure.

Proposition 11.2 (Finite-carrier sorting of the dephased source)**.** *Q-RAIF C's charge-sector sorter [14] applies to L independently realized copies of either ρ_p under its finite-carrier hypotheses: the source marginal is $\rho_p^{\otimes L}$, initially a product with the complete thermal carrier; their external purifications are retained; joint source operations act on both the receiver and its current R pair, while identity on old references refers to systems outside that quartet; the source is neutral in the original zero-mode quartet sector; the original neutral gates and paid loading, unloading and pipeline stages are available; and weighted energy tails control the implementation limit. It returns the source near its blank with entropy slack $o(L)$ and complete output Gibbs deficit $o(V)$, $V = L/\nu$, at*

$$T_{*,p} = T_H + \frac{6\nu h_p}{\pi c_{\mathrm{ex}}}, \qquad c_{\mathrm{ex}} = 4.$$

The same spectral argument applies to a supplied deterministic word with product marginal $\bigotimes_{j=1}^{L} \rho_{p_j}$ and actual entropy $s_L = \sum_j h_{p_j}$. The finite-bank target uses s_L plus the same subextensive margin. If $s_L/L \to s$, its temperature tends to $T_H + 6\nu s/(\pi c_{\mathrm{ex}})$, with no convergence rate required; otherwise there need not be one limiting temperature. This is a block converter, not an adaptive moving apparatus or a per-cycle heat-reading law.

The spectral proof is in Appendix H.1. It uses a supplied product marginal and controlled energy tails, not a local reset certificate for an adaptive history.

12 Relation to Existing Approaches

AdS/CFT [31] and its area–entropy relation [27] supply the dictionary used in Theorem 5.6; it is not derived here or granted to arbitrary selected algebras. MERA [32, 33] implements coarse-graining through disentanglers and isometries, with network depth resembling a radial AdS direction. Different networks can describe the same boundary data. Relating either construction to physical selection requires the explicit algebra-to-geometry map.

Jacobson's horizon Clausius argument [34] shares the entropy-based approach to gravitational dynamics. Section 8 addresses its source-side input: which selecting processes produce the stress whose response is computed. Loop quantum gravity [29, 35] instead quantizes the metric with background independence. Its relation to a path of selected algebras would require a representation map from spin-network states. These connections motivate the proposal without supplying its missing selection law.

13 Conclusion

The gravitational proposal is that physical histories form a localization structure whose slow evolution admits a common geometric law. Modular stability does not select that structure uniquely. A specified half-sided modular family supports conditional holographic matching; the conformal scalar bridge instead relates one prepared source and its Wald surface coefficient to an all-ball test in the same spacetime. Nonzero-frequency preparations can identify the intrinsic response within its stated class. The consistency condition itself still needs a physical explanation.

The source mechanism is explicit. Changed correction, protection and projection interactions change stress. A finite JT mean-action model retains the acquired quantum record, old reference, mechanics, clock, support and recoil, and proves a nonzero record-mediated source return. The spatial return and full-mode quartet bulk strengthen different interfaces. They are evidence for a common theory, not one already joined higher-dimensional apparatus.

Maintenance can favor one selected structure over another. Pointer alignment lowers the complete renewal debit, and a thermal chiral channel of fixed capacity places a quadratic rate price on its export. Matched horizon transport can read that price, provided the actual source, clock reference and heat routing are supplied. Sharp conversion of a fixed product source does not establish sustained renewal or sharp informative adaptive return.

The remaining mechanism is therefore specific: histories must form the accessible sector, the selecting interactions must carry their reaction stress, and one calibrated source must satisfy the same geometric consistency test across those histories. At second order the geometry, propagation, source and region prescription must vary together. A visibility filter cannot replace this dynamics, and changing an algebra at fixed stress cannot create a second first-order sourced Einstein tensor. Establishing this common selection–source–geometry law would turn the adiabatic algebra proposal from an interpretation of supplied gravity into a mechanism for its effective dynamics.

Certificate guide: HAFF-D technical supplement

The main argument and its core results remain in this paper. The complete technical appendices are in `HAFF_D_supplement.pdf`, *Sources, Return and Apparatus Certificates: Technical Supplement to HAFF-D*. This is supplementary documentation for the present paper, not a separate paper. Appendix letters and all existing result and equation numbers are retained. Cross-document references below link to that file; the supplement links back to the main argument.

Main claim or question	Technical certificate	Scope of support
Conditional modular route	Appendix E; Theorem E.5	Definitions and candidate checks, not a physical selection law.
Source changed by actual interaction	Appendix F; Propositions F.1–F.3	Source identification requires the stated reference permissions.
Finite source–response return	Appendices A, G; Propositions A.2, G.7	A fixed-window mean-action return; every source and clock debit remains.
Complete maintenance price	Appendix B; Proposition B.1	Correlations and the complete output deficit enter the export bill.
Horizon transport and clock	Appendix H; Propositions H.1, H.5	A matched source and running-clock reference are required.
Physical preparation and source	Appendix I; Proposition I.2	Finite preparation and the stated material/source matching.
Support, recoil and later reading	Appendix J; Proposition J.6	Retained support and old reference; not a fresh-body replacement.
Spatial and full-mode extensions	Sections J.7, J.8, J.9	Distinct layouts and domains; not one joined apparatus.
Sustained renewal and sharp return	Appendices C, D	Conjecture C.1 remains open; component certificates do not close targets B and C.

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